PROTEOMIC & GENEOMIC DRUG DEVELOPMENT

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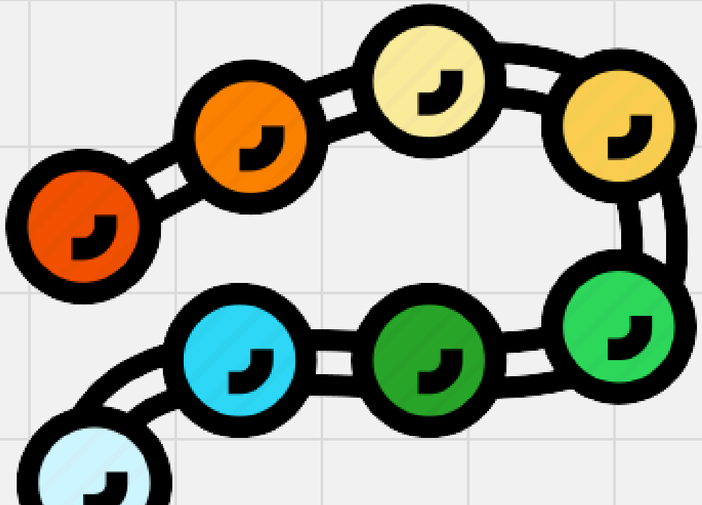
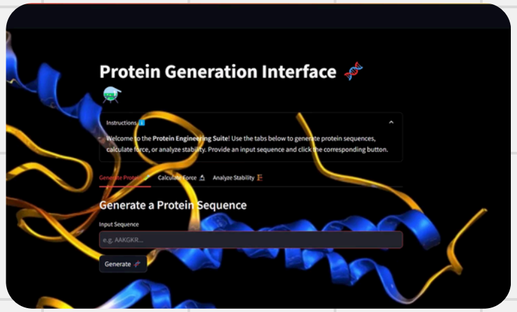
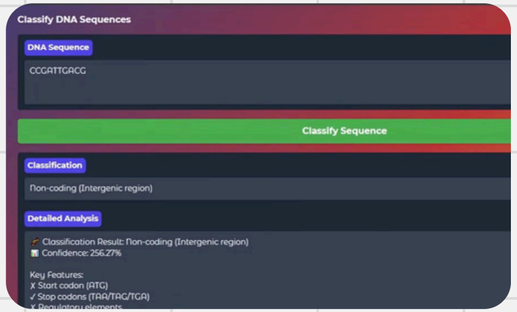
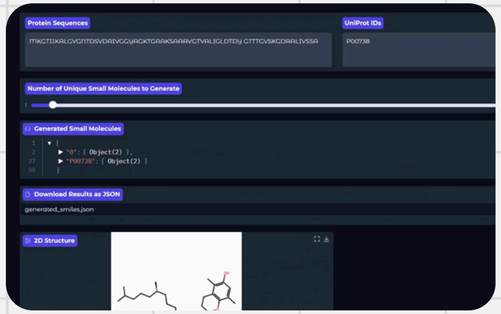
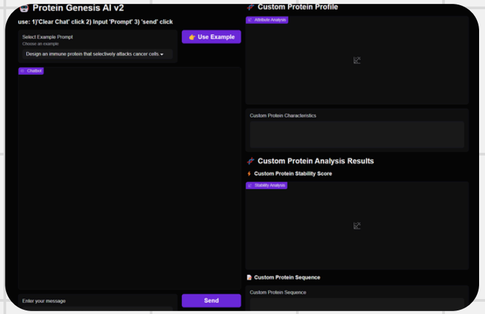


COMING UP









PROTEIN GENERATION & ANALYSIS

STEP 1
CUSTOM MODEL

DNA GENERATION & ANALYSIS

STEP 2
BERT FOR DNA

PERSONALIZED DRUG DEVELOPMENT

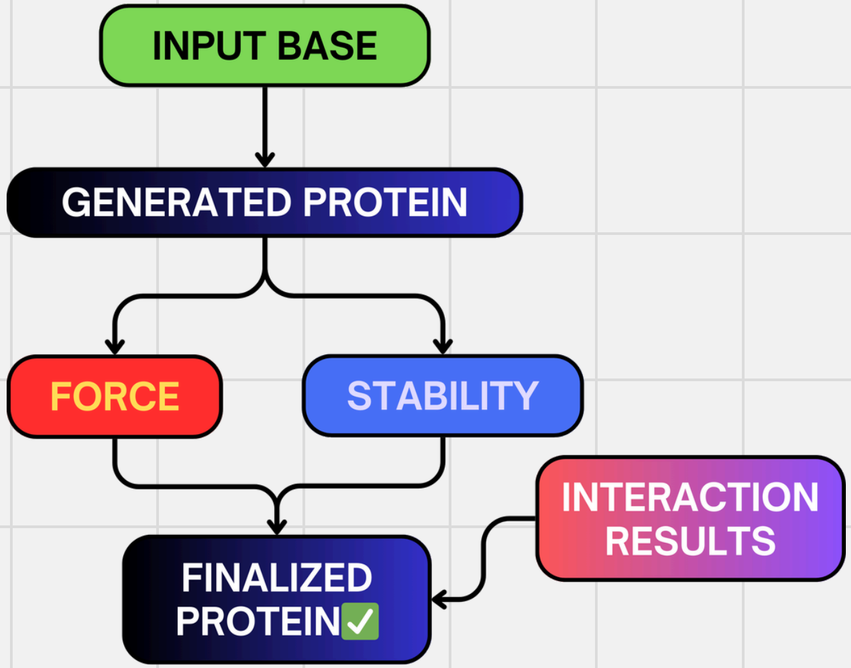
STEP 3
FINE TUNED GPT

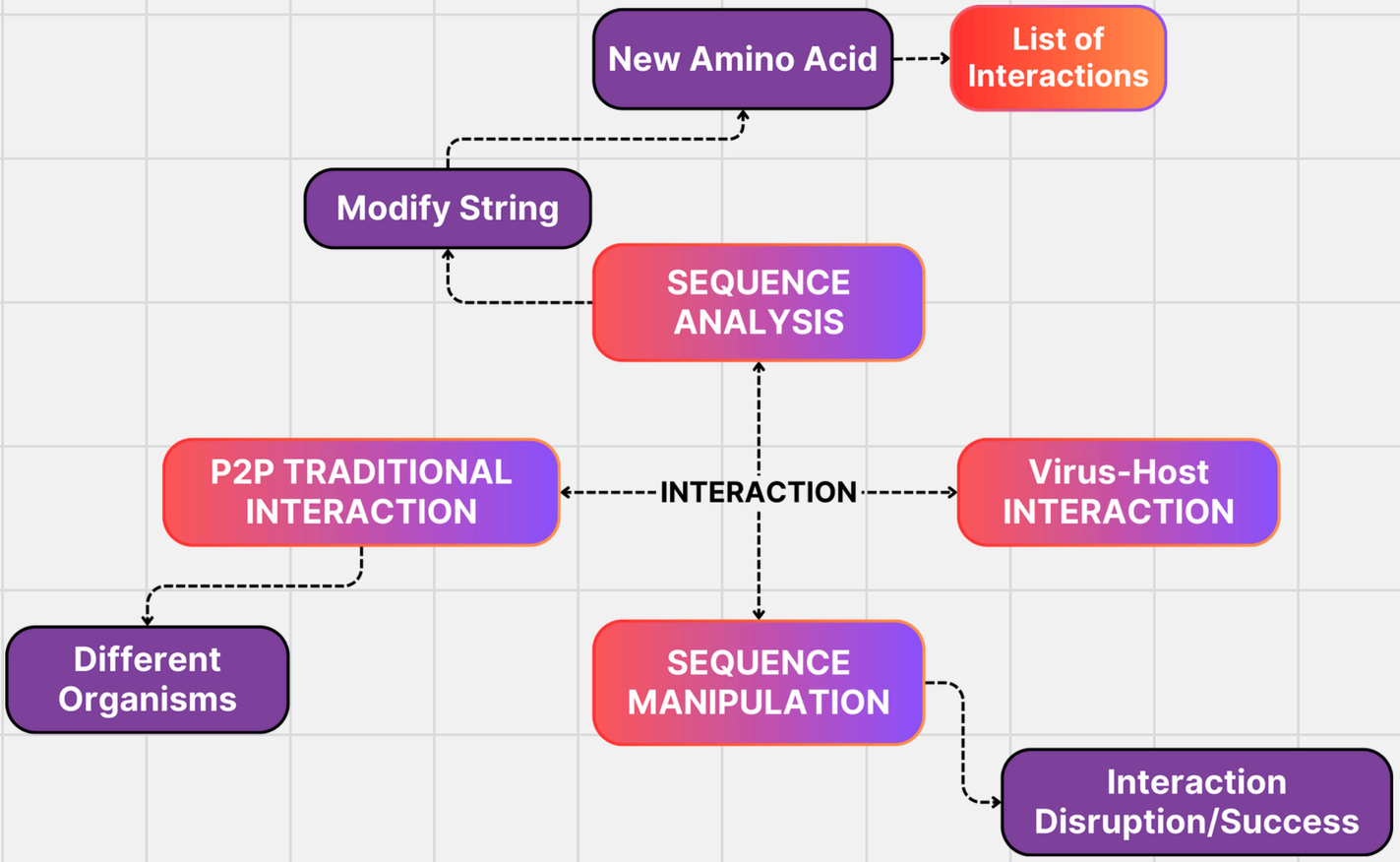
TEXT-GUIDED GENERATION

STEP 4
CUSTOMIZATION USING GENAI

PROTEIN GENERATION & INTERACTION

AIM



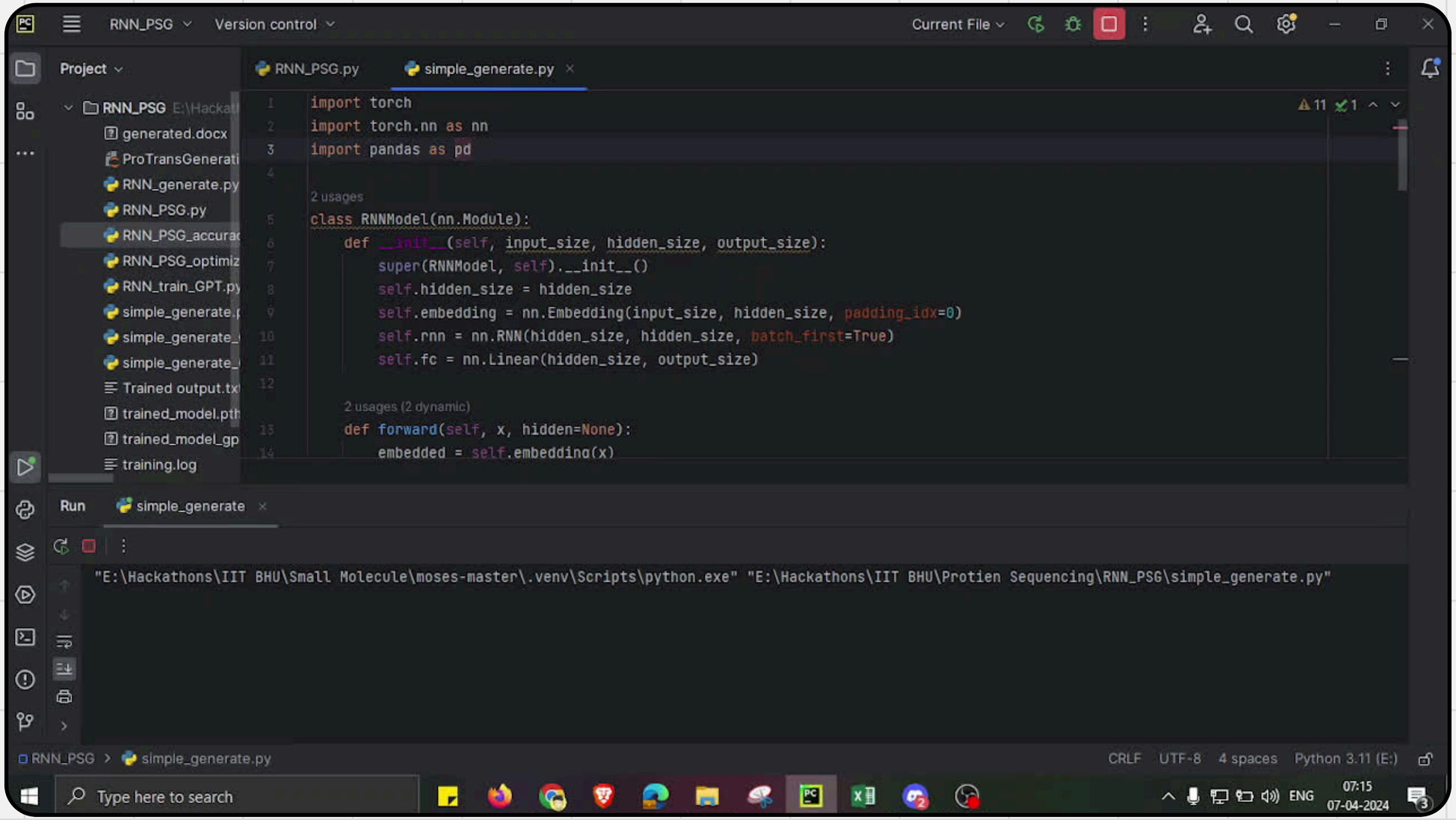


App Flow

CHEMBIOAI

SEQUENCE GENERATION

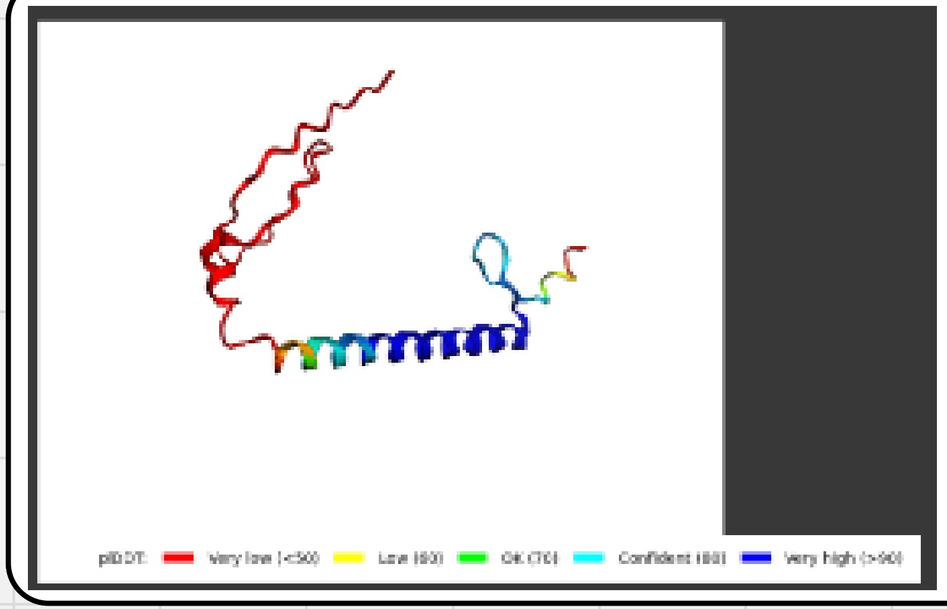
BOLTZMANN

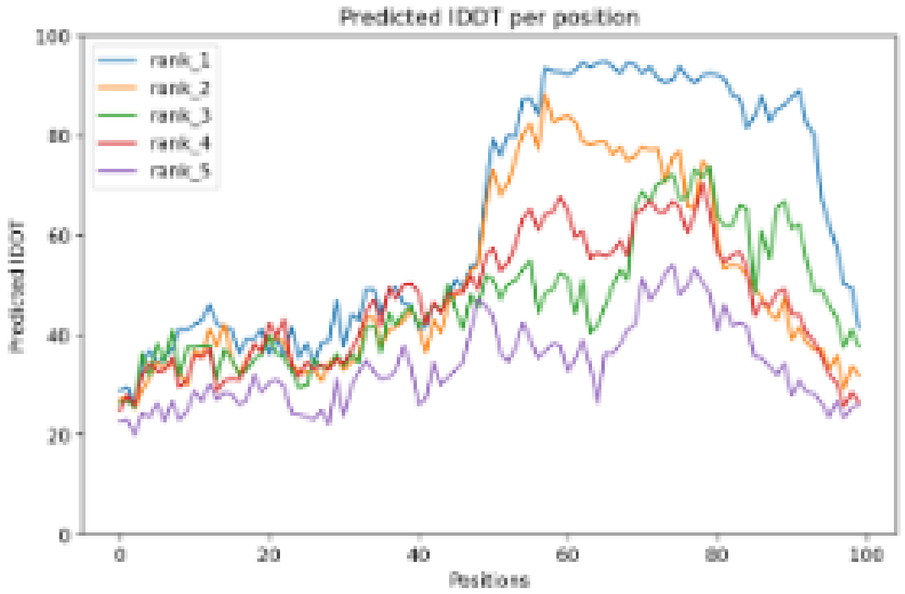


OUTPUT EVALUATION

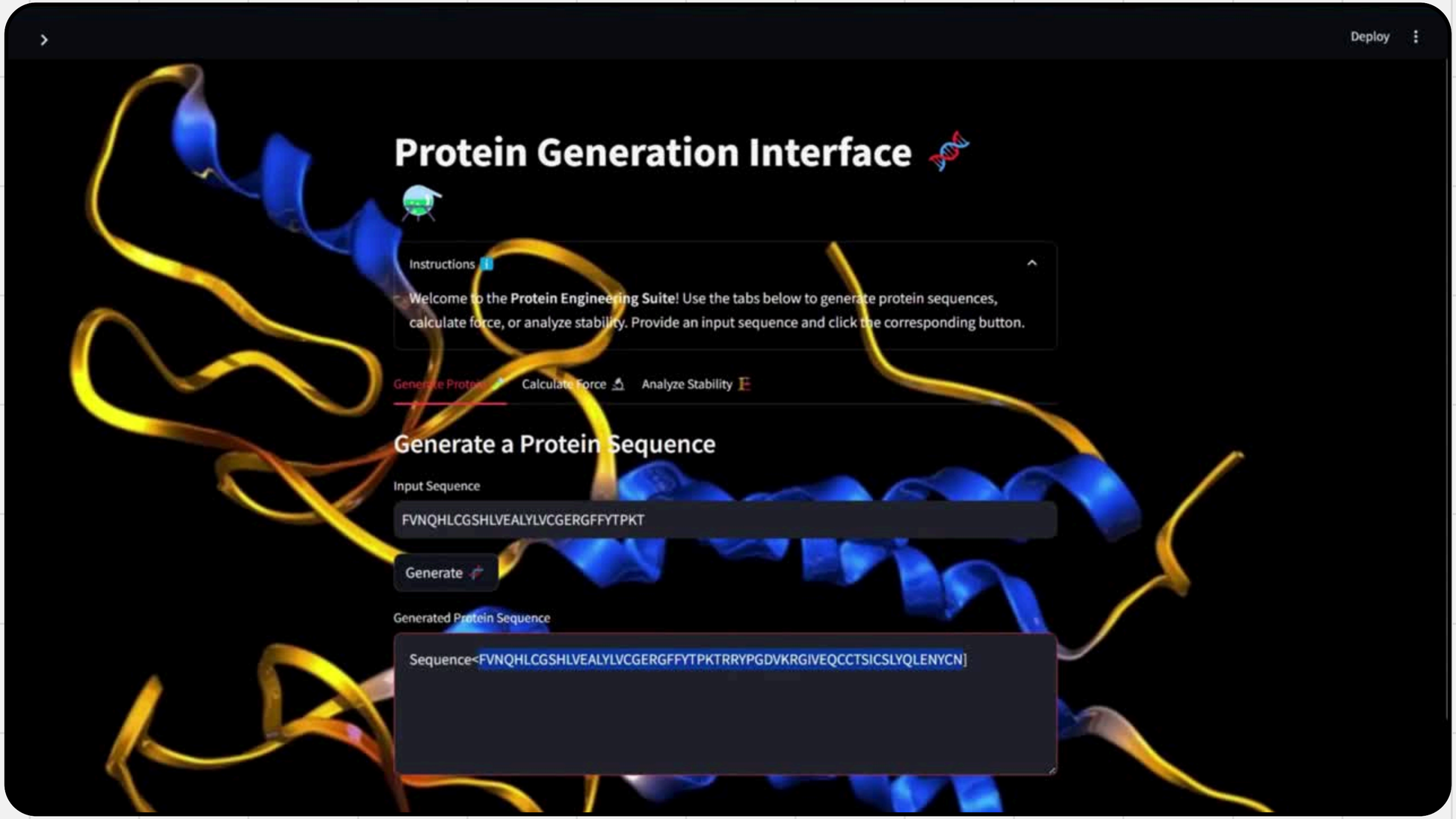
Using AlphaFold2 to Evaluate Generated Protein Sequences

SEQUENCE	MODEL	Predicted local Distance Difference Test (pLDDT)	predicted Template Modeling (pTM)	Tolerence
ACGLESNVGETVIPDGE	ALPHAFOLD2_PTM_MODEL_5_SEED_000:	63.5	0.363	0.582

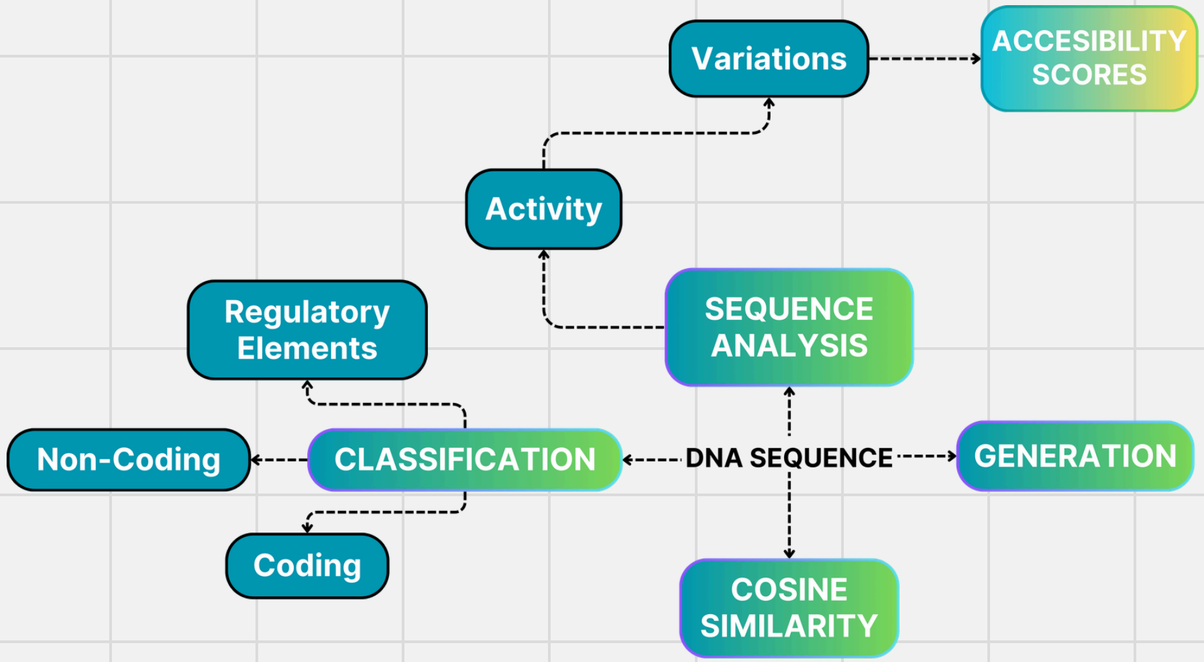




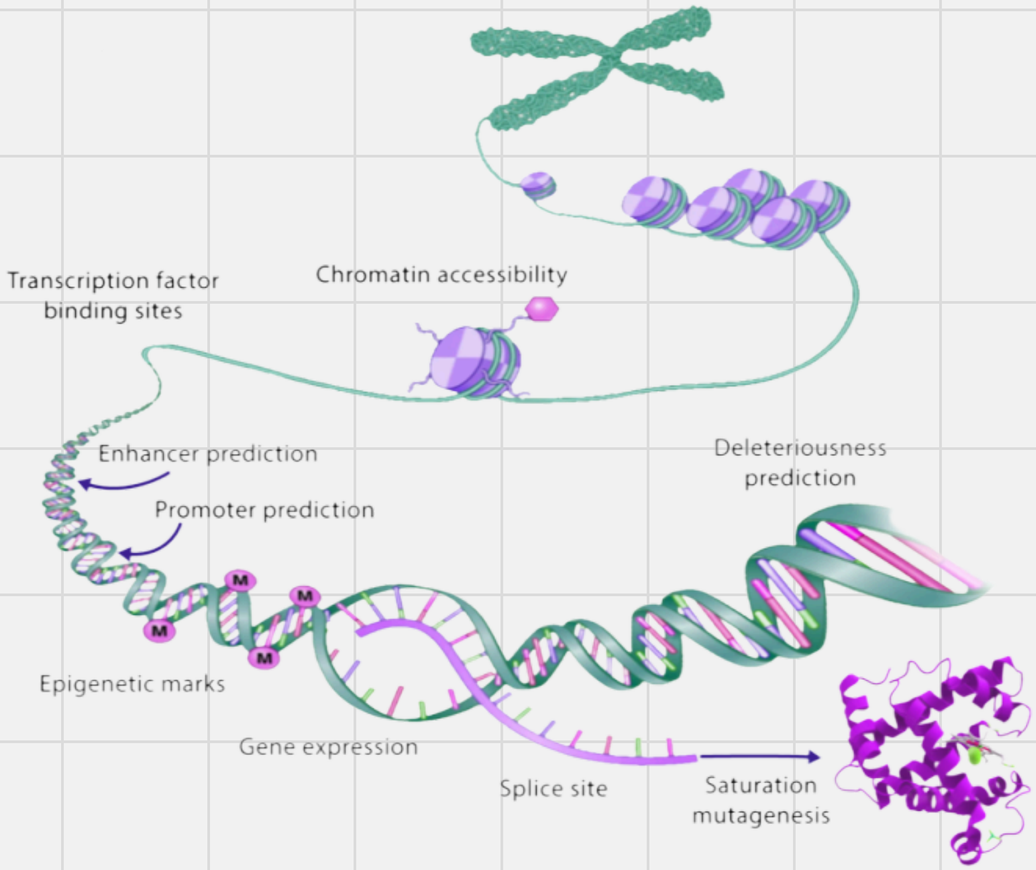
PROTEIN-PROTEIN INTERACTION



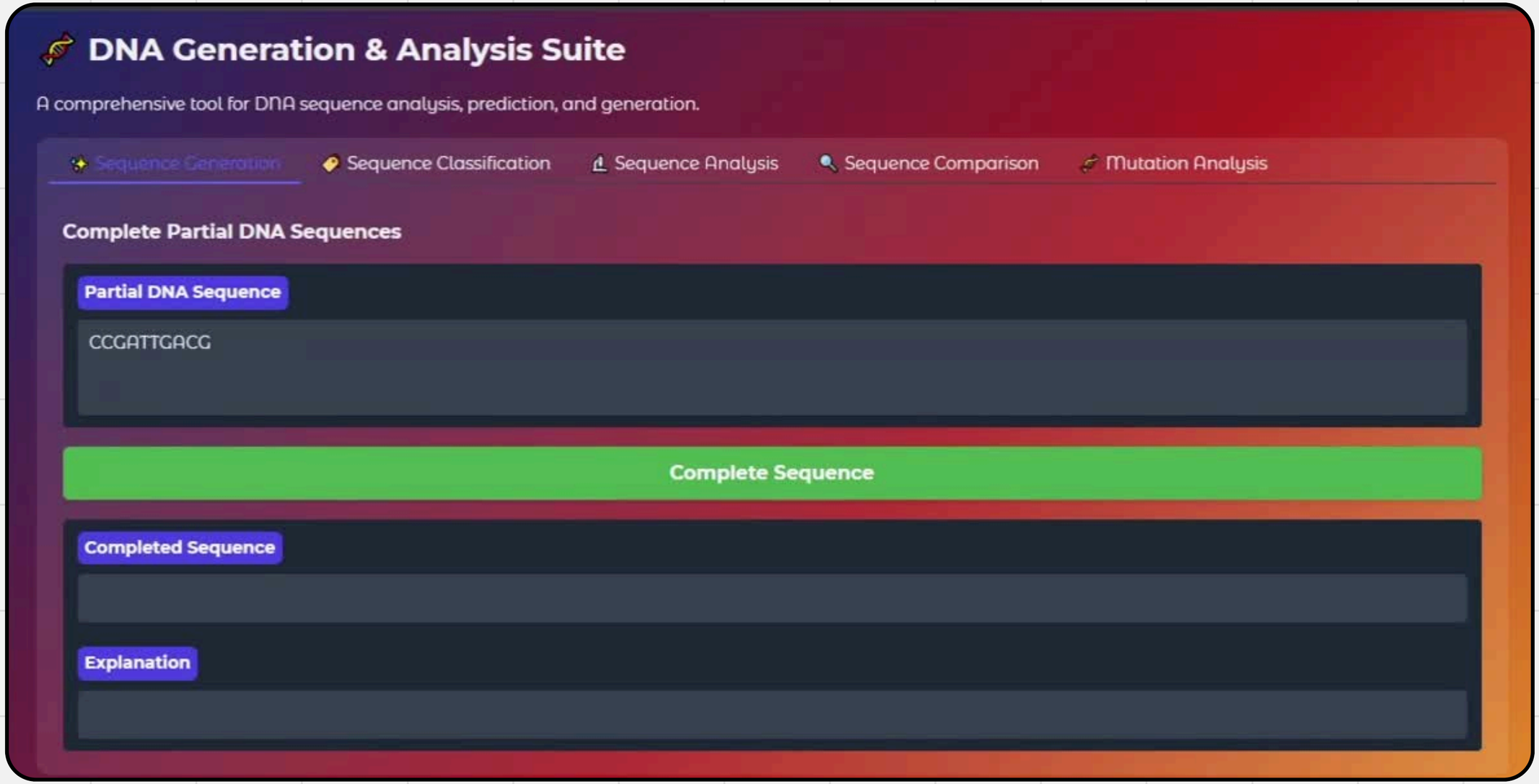
DNA GENERATION & ANALYSIS



App Flow



DNA GENERATION & ANALYSIS



PERSONALIZED DRUG DEVELOPMENT

UniProt

GENERATIVE PRE-TRAINED TRANSFORMER

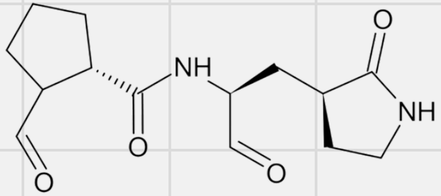
SUPERVISED TRAINING

POLICY OPTIMIZATION

DRUG GENERATOR

FEEDBACK

- REPEATED PENALTY
- RELAVANCY TRACKER
- BONDAGE PROBABILITY



DRUG-GEN DEVELOPMENT

PROTEIN SEQUENCE/UNIPROT ID

DRUG GENERATOR

SMILES SEQUENCE

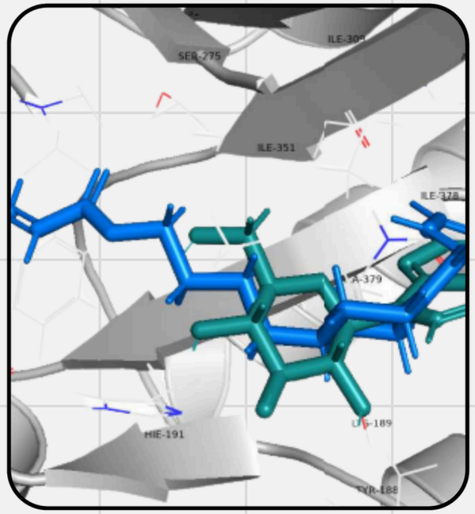
VALIDITY

DIVERSITY

NOVELTY

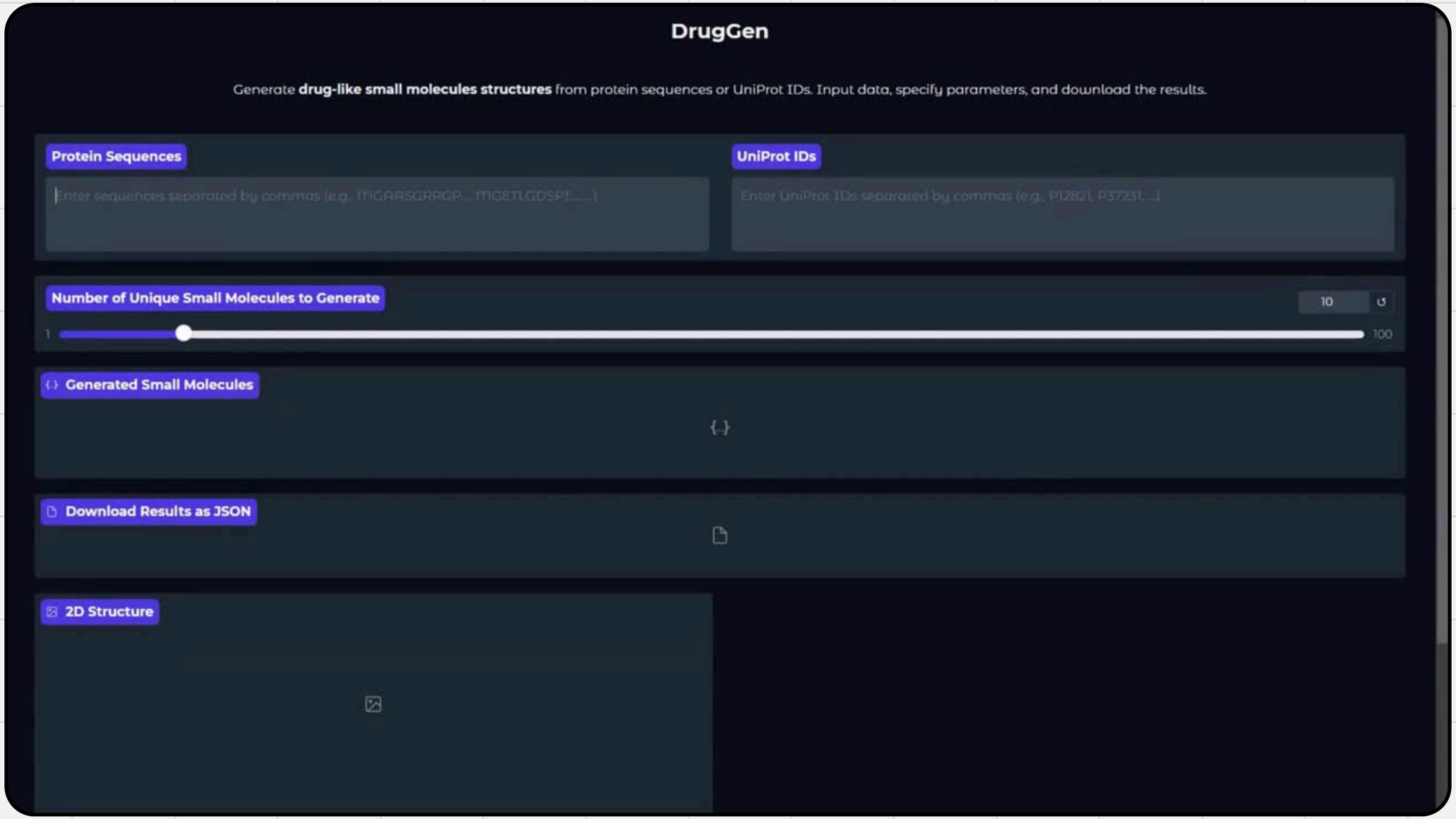
BINDING AFFINITY

DOCKING SIMULATION



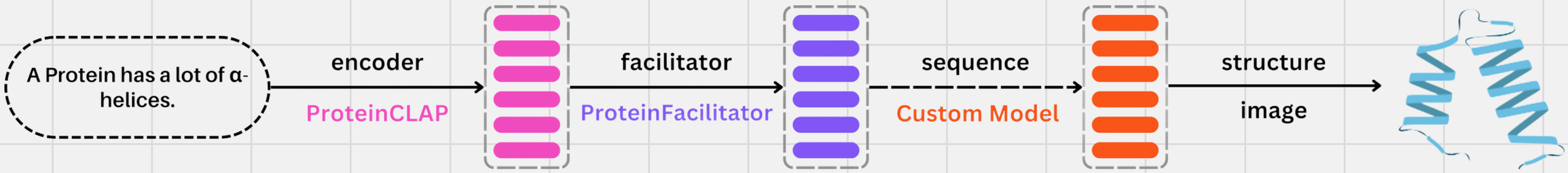
APP-FLOW

DRUG DEVELOPMENT DEMONSTRATION

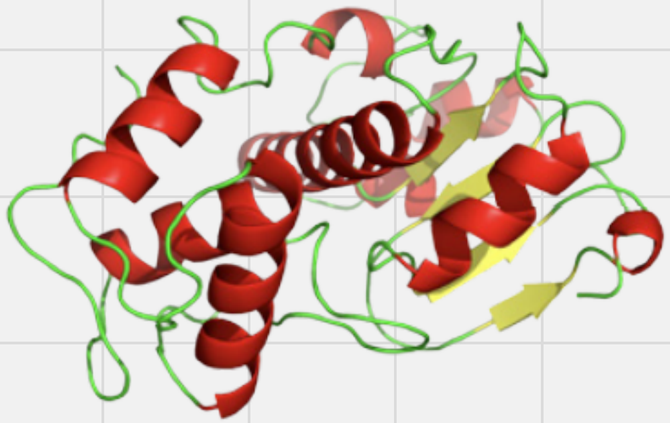


TEXT GUIDED PROTEIN GENERATION

PIPELINING TEXT TO PROTEIN IMAGE GENERATION MODEL

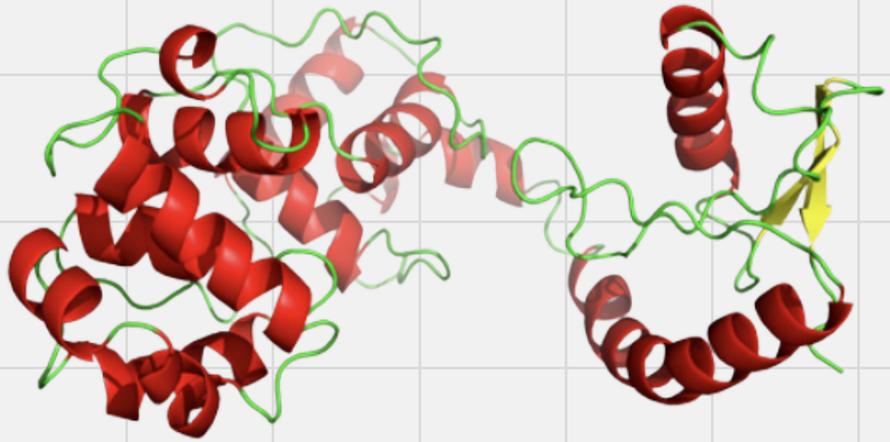


RESULTS

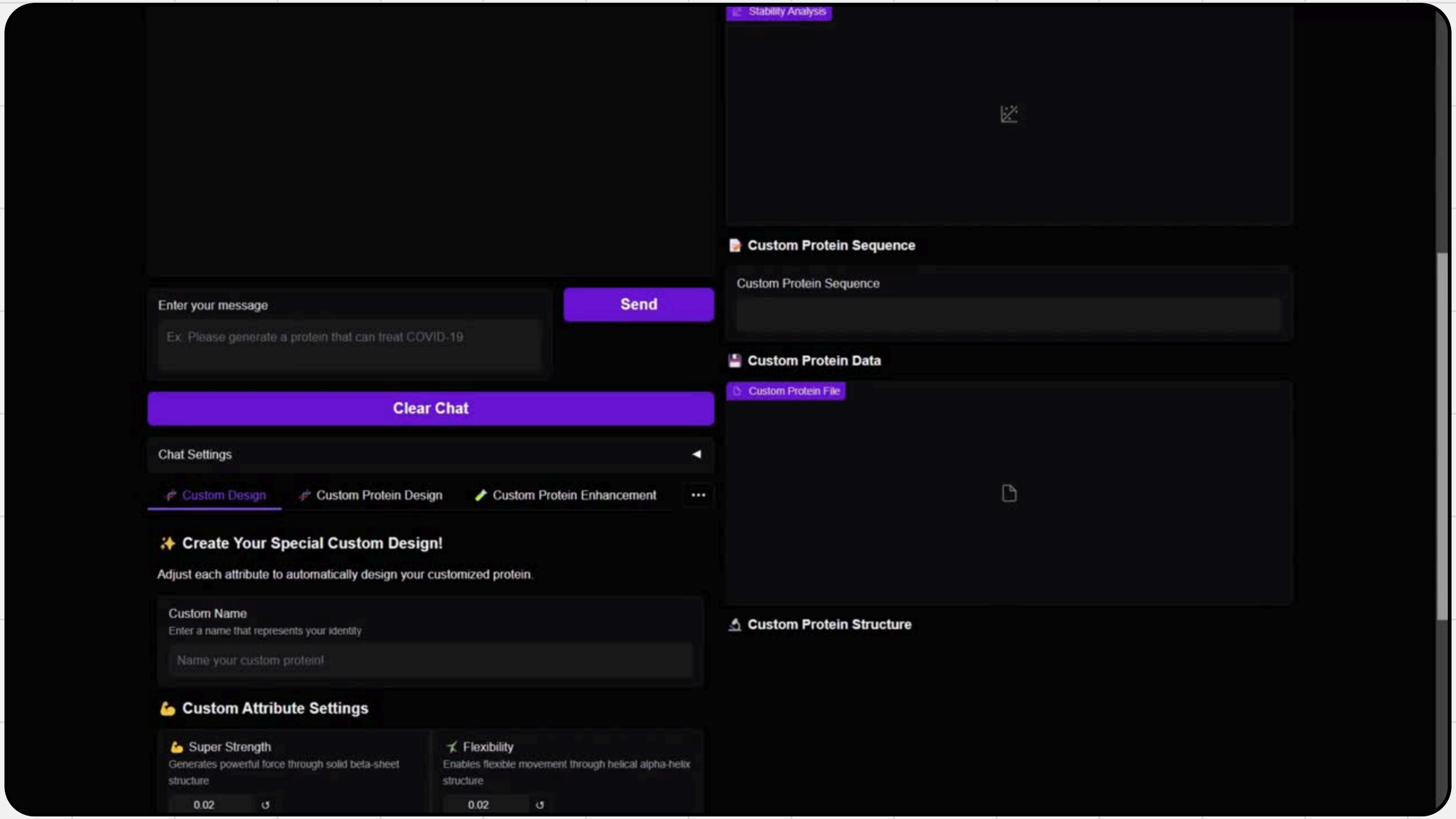


NORMAL

OUR RESULT



TEXT GUIDED PROTEIN GENERATION



THANK YOU

WE ARE OPEN TO QNA

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